

Yuanqing Wang

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EDUCATION

- **The University of Texas at Dallas** Richardson, TX
Ph.D. Student in Mathematics; Advisors: Prof. Baris Coskunuzer (Math), Prof. Yapeng Tian (CS) *Aug. 2024 – Present*
 - GPA: 3.77/4.00.
 - Selected Coursework: Numerical Linear Algebra (A), Deep Learning (A).
- **Capital Normal University** Beijing, China
M.S. in Mathematics; GPA: 3.86/4.00 *Sep. 2021 – Jun. 2024*
 - Thesis: Calabi Conjecture and Kähler-Einstein Metric. Graduate training in differential geometry, topology, and analysis.
- **Xiamen University** Xiamen, China
B.S. in Computational Mathematics; Elite Students' Program *Sep. 2016 – Jun. 2021*
 - Elite Students' Scholarship. China Undergraduate Mathematical Contest in Modeling, 1st Prize in Fujian Province (2018).

RESEARCH INTERESTS

My research focuses on post-training for long-horizon LLM agents, particularly agentic reinforcement learning, credit assignment, and self-evolving agents.

SELECTED RESEARCH

- **dwDPO: Training Multi-Turn LLM Agents via Divergence-Weighted Direct Preference Optimization** *EMNLP 2026 submission*
 - Proposed divergence-weighted DPO for multi-turn credit assignment; outperformed standard DPO on all 6 long-horizon benchmarks, improving average accuracy from 35.9 to 37.0 at no added inference cost.
 - **Contribution:** Co-first author; conceived the entire method, developed all mathematical formulations and theoretical analysis, and led and coordinated collaborators in executing the experiments.
- **Skill-CDPO: Evolving Agent Tool-Use via Critical Step Preference Optimization** *EMNLP 2026 submission*
 - Introduced Critical Step DPO (CDPO), which identifies capability bottleneck steps via rollout divergence and constructs distributional preference pairs weighted by step-level criticality and pair-level score gaps. 8B model matched or exceeded GPT-5.2 on medical agent benchmarks.
 - **Contribution:** Designed the CDPO method, mathematical formulations, and theoretical analysis.
- **AVSG: Audio–Visual Scene Graphs Measure What Omni-Modal Models Actually Hear** *ACL ARR, Aug. 2026*
 - Introduced a 1,024-clip audio–visual scene graph benchmark; across 12 model configurations, frontier models scored 0.67–0.76 on audio–audio relations versus ≤ 0.21 for open models.
 - **Contribution:** First author; conceived the benchmark and designed the annotation schema and rubric; built the data-curation and AI-prefill pipeline using Qwen3 and the Gemini API, together with a Gradio interface for human verification.
 - Recruited, organized, and led 7 undergraduate annotators through dataset construction and wrote the paper.
- **Heat Field Signatures: Density-Aware Geometry for Point Cloud Classification** *To be submitted to ICLR 2027*
 - Introduced Heat Field Signatures for point-cloud classification; achieved 59.6% versus 35.7% for the best deep baseline on protein classification while running roughly $9\times$ faster end-to-end.
 - **Contribution:** First author; implemented the proposed method, selected the evaluation datasets, discovered its rotation-invariance property, ran all experiments, analyzed the results, and drafted the experimental section of the paper.
- **When Do Critics Beat Groups? A Crossover Rule for Credit Assignment in Long-Horizon Agents** *In progress; ICLR 2027 target*
 - Developing CROC, a two-timescale advantage estimator using a boundary critic for across-turn credit and group-relative estimation within turns; derived a crossover rule predicting when critic-based credit should outperform group baselines.
 - **Contribution:** Conceived the CROC method and crossover-rule formulation; developing the theoretical framework, implementation, and experimental evaluation for long-horizon agent RL.

EARLIER RESEARCH

- **Solving Differential Equations Using Artificial Neural Networks** Xiamen University
National Undergraduate Innovation Project (Lead); Advisor: Prof. Chuanju Xu *Mar. 2019 – Apr. 2020*
 - Led a national-level undergraduate innovation project. Implemented MLP-based neural network solvers for PDEs with prescribed boundary conditions, among the early efforts applying neural networks to PDE problems.

INDUSTRY EXPERIENCE

- **Gathssen Investment Co., Ltd.** Shenzhen, China
Quantitative Strategy Developer Intern *Jun. 2021 – Aug. 2021*
 - Built a backtesting framework for dollar-cost averaging strategies and benchmarked multiple variants. Contributed to the development of a cash-and-carry arbitrage strategy.
 - Independently developed a cointegration-based cross-asset hedging strategy with full backtesting pipeline, from research to simulation.

TECHNICAL SKILLS

- **LLM Post-Training:** verl (GRPO, DPO), preference optimization, rollout and trajectory generation, benchmark design and evaluation
- **LLM Systems:** PyTorch, Hugging Face Transformers, vLLM, multimodal model deployment, Gemini API
- **Systems & Infrastructure:** Linux, Git, Slurm, Apptainer, H100/H200/GH200 GPU clusters
- **Programming & Tools:** Python, NumPy, pandas, Gradio

HONORS & AWARDS

- **NSM Summer Research Fellowship, UT Dallas** Summer 2026
- **NSM McDermott PhD Admission Fellowship, UT Dallas** Sep. 2024
- **Excellent Academic Scholarship, Capital Normal University** Dec. 2022 – Dec. 2023
- **China Undergraduate Mathematical Contest in Modeling, 1st Prize (Fujian Province)** Sep. 2018
- **Elite Students' Scholarship, Xiamen University** Sep. 2018 – Sep. 2019

TEACHING

- **Teaching Assistant, UT Dallas — Linear Algebra, Differential Calculus** 2025 – 2026
- **Teaching Assistant, Capital Normal University — Real Analysis, Complex Analysis** 2023 – 2024